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Keywords:

C, C++, Python
Embedded, Machine
Learning, Git

Hean Loong Ong

Staff Software Engineer
Penang, Malaysia

Summary

Experienced Software Engineer with 20 years of development in Embedded and High Speed Networks

Experience

Lattice Semiconductor

Staff Software Engineer (Customer Application)

November 2023 - Present (2 year 5 months)

Penang, Malaysia

Tech Stack

- **FPGAs:** Lattice CPNX, Lattice Avant-E
- **Processors:** RISC-V (100 MHz soft-core)
- **Operating Systems:** FreeRTOS, Zephyr OS
- **Protocols:** Industrial Ethernet (EtherCAT, PROFINET)
- **Optimization/Hardware:** FPGA system cache, Memory Management

Key Achievements

- **Technical Leadership:** Served as Technical Team Lead for an Industrial Ethernet group, defining development strategies and managing a three-person engineering team.
- **Ecosystem Enablement:** Successfully ported and enabled EtherCAT and PROFINET applications on FreeRTOS for 100 MHz RISC-V cores within the Lattice CPNX architecture.
- **Reference Design Development:** Led the implementation of embedded reference designs for Zephyr OS across both CPNX and Avant-E FPGA platforms.
- **Performance Optimization:** Significantly improved FreeRTOS network responsiveness, achieving a latency of approximately 200 μ s per ping cycle.
- **Hardware-Level Performance Tuning:** Enhanced system timing and reliability for industrial use cases by implementing memory management utilizing the FPGA system cache.

Infoblox

Staff Software Engineer (R&D)

July 2020 - November 2023 (3 years 5 months)

Selangor, Malaysia

Tech Stack

- **Networking & Performance:** Intel DPDK, HTTP/2.0, TCP/IP, DNS, Packet Sniffing, Intel DPDK drivers.
- **Security:** DDoS Mitigation, Packet Inspection Engine, DNS Packet Filtering.
- **Languages & Tools:** Python (2.7.0, 3.8), Linux, Security Rule Parsing Engines.
- **Virtualization:** KVM, VMware, Virtual Machine (VM) performance tuning.

Key Achievements

- **DDoS Mitigation & Infrastructure Hardening:** Integrated Intel DPDK software drivers with a high-performance packet inspection engine on Linux to shield DNS servers from malicious DDoS attacks.
- **Advanced Threat Detection:** Developed an HTTP/2.0 backend security engine for DNS, enabling real-time packet sniffing to detect and block malware from communicating with Command & Control (C&C) servers.
- **Security Engine Engineering:** Designed and maintained a robust Python-based rule parsing engine on Linux, automating DNS packet filtering and database security management.
- **Virtualization Performance Optimization:** Diagnosed and resolved critical packet processing bottlenecks and intermittent reset issues on KVM and VMware platforms, ensuring stable performance in virtualized environments.
- **Legacy System Modernization:** Led a strategic technical migration of the legacy Python engine from version 2.7.0 to 3.8

Intel Corporation

Senior Software Engineer (R&D)

September 2016- July 2020 (3 years 10 months)

Penang, Malaysia

Tech Stack

- **Silicon & Architectures:** Intel SoC FPGAs (Stratix 10, Arria 10, Agilex), ARM Cortex-A53/A55/A9 Hard Processor Systems (HPS).
- **Operating Systems & Build Tools:** Linux Kernel (Kernel-level debugging and porting), Yocto Project, Poky.
- **Drivers & Subsystems:** Linux DRM/KMS (DisplayPort), FPGA Manager, I2C, SPI, GPIO, Ethernet, PCIe.
- **DevOps & Security:** CI/CD (Yocto-based pipelines), Secure Software Development Cycle (SSDLC), Git/Gerrit.

- **AI/ML Validation:** KNN (K-Nearest Neighbors), SVM (Support Vector Machine), Intel Machine Learning IP.

Key Achievements

- **Silicon Enablement & Driver Development:** Spearheaded the Linux enablement for high-performance Intel SoC FPGAs (Agilex, Stratix 10, and Arria 10), involving deep-level kernel porting and the development of robust drivers to bridge the Hard Processor System (HPS) and FPGA fabric.
- **Reference Design Strategy:** Directed the technical strategy for the Intel SoC FPGA Golden Reference Design (GHRD), ensuring a stable and optimized hardware-to-software handoff for enterprise-grade silicon.
- **Embedded Video Subsystems:** Implemented embedded Linux drivers for DisplayPort on Intel SoC FPGAs, resolving complex hardware-timing issues and ensuring high-bandwidth video data path stability.
- **Automated Infrastructure:** Engineered a scalable DevOps framework for FPGA software development, utilizing Yocto-based CI/CD pipelines to automate cross-compilation, kernel builds, and hardware validation tests.
- **Security Leadership:** Served as the Technical Team Lead for the Secure Software Development Cycle (SSDLC), implementing hardening protocols for Intel SoC FPGA Linux software to mitigate system-level vulnerabilities.
- **Machine Learning Validation:** Developed and integrated specialized validation modules for Intel Machine Learning IPs, optimizing the performance of KNN and SVM algorithms on FPGA-based hardware accelerators.

Infoblox

Senior Software Engineer (R&D)

March 2015- July 2016 (1 year 5 months)

Selangor, Malaysia

Tech Stack

- **Networking & Hardware:** 10GbE, PCIe Gen3, Intel DPDK, DMA (Scatter-Gather), MSI-X, PCIe Endpoint design.
- **Systems & Architecture:** Linux Kernel, Bare-metal, MIPS (Multi-core), KVM/VMware.
- **Languages & Protocols:** C, MIPS Assembly, Python, DNS, IPv6, BGP, Zebra/Quagga.

Key Achievements

- **High-Speed Network Security:** Designed 10G DDoS mitigation and DNS packet inspection engines by offloading processing to hardware via custom PCIe drivers.

- **Embedded Firmware:** Developed mission-critical bare-metal firmware for MIPS-based 10G PCIe endpoint devices, ensuring high-reliability host-to-device communication.
- **Protocol Modernization:** Ported Zebra software to custom Linux platforms to enable high-performance IPv6 BGP routing.
- **PCIe Performance Optimization:** * Saturated 10G links using **Scatter-Gather DMA** and optimized payload sizes (MPS/MRRS).

Motorola Solutions

Senior Software Engineer (R&D)

October 2008- February 2015 (6 year 5 months)

Penang, Malaysia

Tech Stack

- **Languages:** C, C++
- **Operating Systems:** Embedded Linux, Monta Vista Real-Time Linux, Nucleus OS (Legacy RTOS)
- **Hardware & Domain:** Motorola Radio (PCR, TETRA), Digital Signal Processors (DSP), Hardware-Software Interfacing
- **Architectures & Tools:** UML Framework, DSP Frame Programming, Low-level Hardware Abstraction Layers (HAL)
- **Methodology:** Agile Development

Key Achievements

- **SME & Architectural Design:** Served as the Subject Matter Expert for Motorola Radio PCR and TETRA accessory software, defining the embedded Linux architecture for next-generation radio accessories.
- **DSP Interface Engineering:** Maintained and optimized a critical C++ codebase utilizing a UML framework to manage low-level hardware DSP interfaces and frame programming.
- **OS Platform Migration:** Successfully led the complex migration of legacy radio codebase from Nucleus OS to the Monta Vista Real-Time Linux platform, modernizing the system's multitasking and networking capabilities.
- **Technical Leadership:** Directed a two-person Agile team through full-cycle software development, focusing on high-priority debugging and the implementation of low-level embedded features.

Universiti Sains Malaysia

Researcher (R&D)

July 2005- July 2008 (3 years)

Penang, Malaysia

Tech Stack

- **Protocols:** IPv6, IPv4, BGP, TCP/IP, IPsec.
- **Operating Systems:** OpenBSD, Red Hat Enterprise Linux (RHEL), Linux Kernel.
- **Networking & Hardware:** Satellite Communication (USM-KEIO), Open Source Switch Emulators (IMUNES, CORE), KAME stack.
- **Languages & Tools:** C (Kernel-level programming), Shell Scripting, sysctl, modprobe, Kernel patching tools.

Key Achievements

- **Educational Infrastructure Development:** Successfully enabled and configured an open-source switch emulator to support large-scale training and experimental IPv6 network research, providing a sandbox for testing next-generation protocols.
- **International Research Collaboration:** Maintained the technical operations of the USM-KEIO satellite communication project, optimizing high-latency data transmission and connectivity over IPv6-enabled satellite links.
- **Kernel-Level Enablement:** Authored and integrated critical patches for IPv6 network kernel drivers on OpenBSD and Red Hat platforms, ensuring RFC-compliant packet handling and resolving driver-level bottlenecks for production environments.

Education

Universiti Sains Malaysia (2005 - 2008)

Masters in Computer Science (Computer Networks)

Universiti Tunku Abdul Rahman (2003 - 2005)

Bachelor in Computer Science

Kolej Tunku Abdul Rahman (2001 - 2003)

Diploma in Computer Science